REQUIREMENT DOCUMENT

Project Name:

Cybersecurity Incident Response Dashboard for SMEs

Company:

CyberShield Solutions

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Academic Year:

2026–2028

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INTRODUCTION

1. Introduction

In today’s digital era, cybersecurity has become a critical concern for all organizations, particularly Small and Medium Enterprises (SMEs). As businesses increasingly rely on digital systems, cloud platforms, and online transactions, the attack surface for cybercriminals has expanded significantly. SMEs are especially vulnerable because they often lack advanced security infrastructure and dedicated cybersecurity teams.

Modern cyber threats have evolved in both complexity and frequency. Attackers now use automated tools, artificial intelligence, and social engineering techniques to exploit system vulnerabilities. Common attacks include ransomware, phishing, malware infections, insider threats, credential theft, and distributed denial-of-service (DDoS) attacks.

Recent global cybersecurity incidents demonstrate the severity of these threats. Ransomware attacks have caused entire organizations to halt operations, sometimes demanding large financial payments for data recovery. Phishing attacks continue to deceive employees into revealing sensitive credentials. Data breaches expose confidential customer and business data, leading to legal consequences and loss of trust.

Despite these risks, many SMEs still rely on outdated or manual methods such as spreadsheets, emails, or disconnected tools to manage security incidents. This leads to delayed response times, miscommunication between teams, lack of accountability, and incomplete incident tracking.

The absence of a structured incident response system makes it difficult for organizations to identify, prioritize, and resolve threats efficiently. As a result, cyber incidents often escalate, causing operational disruption and financial loss.

This project proposes a **Cybersecurity Incident Response Dashboard**, designed to centralize and automate incident management. The system aims to improve visibility, enhance coordination, and provide real-time insights into cybersecurity threats, enabling SMEs to respond quickly and effectively to cyber incidents.

PROJECT OVERVIEW

2. Project Overview

The Cybersecurity Incident Response Dashboard is a centralized digital platform designed to manage cybersecurity incidents in real time. It integrates detection, monitoring, classification, tracking, and response functionalities into a single system.

The dashboard acts as a control center for cybersecurity operations, allowing IT teams to monitor ongoing threats, analyze incident severity, and coordinate responses efficiently.

Key Objectives:

- To provide real-time monitoring of cybersecurity incidents across systems and networks
- To reduce incident response time by enabling immediate alerting and escalation
- To improve collaboration between security teams through centralized communication
- To prioritize incidents based on risk level, impact, and urgency
- To generate analytical insights for long-term security improvement
- To create a structured workflow for incident handling and resolution

System Importance:

A centralized dashboard eliminates fragmentation in cybersecurity management. Instead of relying on multiple tools or manual reporting, organizations gain a unified view of their security posture. This improves decision-making and ensures consistent response procedures.

Expected Outcome:

The expected outcome is a fully functional conceptual model (or prototype) of a dashboard that enhances cybersecurity readiness and supports SMEs in managing digital threats effectively.

PROBLEM STATEMENT

3. Problem Statement

Small and Medium Enterprises face increasing cybersecurity challenges that significantly impact their operational stability and financial security. Despite the growing threat landscape, many SMEs lack structured systems for managing and responding to cyber incidents.

Key Problems:

- **Lack of centralized incident management:** Security events are often tracked using separate tools, spreadsheets, or emails, leading to fragmented information.

- **Slow response times:** Without real-time alerts and structured workflows, organizations take longer to detect and respond to attacks.
- **Poor prioritization of threats:** All incidents are not treated with appropriate urgency, causing critical threats to be overlooked.
- **Limited cybersecurity expertise:** SMEs often do not have dedicated cybersecurity teams or trained professionals.
- **Inadequate documentation:** Incident records are incomplete or inconsistent, making it difficult to analyze past attacks.

Impact of These Problems:

- Financial losses due to downtime and data breaches
- Damage to brand reputation and customer trust
- Legal and regulatory consequences
- Increased vulnerability to repeated attacks

These issues highlight the urgent need for a structured and automated incident response system.

TARGET USERS

4. Target Users

The system is designed for multiple user groups within an organization, each having different responsibilities and levels of access.

User Type	Role Description
IT Administrators	Responsible for system monitoring, maintenance, and incident resolution coordination
Security Analysts	Analyze threats, investigate incidents, and identify attack patterns
Managers	Make strategic decisions based on security reports and risk analysis
Compliance Officers	Ensure adherence to cybersecurity regulations and audit requirements
Employees	Report suspicious activities and follow security guidelines

User Interaction:

Each user interacts with the dashboard differently. For example, analysts focus on detailed incident data, while managers rely on summarized reports and visual dashboards for decision-making.

USER NEEDS

5. User Needs

To be effective, the system must address both the technical and operational needs of users.

Functional Needs:

- Real-time detection and notification of security incidents
- Centralized dashboard for monitoring all threats
- Ability to assign incidents to specific team members
- Automated severity classification system
- Incident history and reporting capabilities

Non-Functional Needs:

- User-friendly interface for both technical and non-technical users
- High system reliability and uptime
- Scalability to handle increasing security events
- Data security and access control mechanisms

Operational Needs:

- Faster coordination between departments
- Clear visibility into incident progress
- Reduction in manual workload

PROPOSED SOLUTION

6. Proposed Solution

The proposed system is a centralized Cybersecurity Incident Response Dashboard that integrates all cybersecurity operations into a unified platform.

Core Capabilities:

- Real-time monitoring and alert generation
- Centralized incident logging system

- Severity-based prioritization and classification
- Automated workflow for incident response
- Visual analytics and reporting tools

Working Mechanism:

When a security threat is detected, the system generates an alert and automatically categorizes it based on severity. The incident is then assigned to the appropriate team member, who follows a structured response workflow until resolution.

Benefits of the Solution:

- Reduces response time significantly
- Improves communication between teams
- Enhances decision-making through data visualization
- Provides structured documentation of all incidents

CORE FEATURES (PART 1)

7. Core Features

7.1 Incident Overview

The incident overview section provides a real-time snapshot of all ongoing security events within the organization.

It categorizes incidents into active, pending, and resolved states, allowing users to quickly assess system health and security posture.

This feature helps organizations identify critical issues at a glance and ensures that no incident goes unnoticed.

7.2 Alert System

The alert system is responsible for generating real-time notifications when suspicious activity is detected.

Alerts are classified based on severity levels:

- Low: Minor issues requiring monitoring
- Medium: Potential threats requiring attention
- High: Serious threats requiring immediate action
- Critical: Severe attacks requiring urgent response

This ensures proper prioritization and reduces response delays.

CORE FEATURES (PART 2)

7.3 Incident Tracking

Incident tracking provides a complete lifecycle view of each cybersecurity event from detection to closure.

It records all actions taken, assigned personnel, timestamps, and resolution steps.

This improves accountability, transparency, and helps in post-incident analysis.

7.4 Risk Awareness

The risk awareness module evaluates threats based on likelihood and potential impact.

It helps organizations identify which threats pose the highest risk and should be addressed first.

Common threats include ransomware, phishing attacks, malware infections, insider threats, and data breaches.

CORE FEATURES (PART 3)

7.5 Response Guidance

This feature provides structured step-by-step instructions for handling incidents.

Typical response stages include detection, classification, assignment, investigation, containment, recovery, and closure.

This ensures consistency in handling incidents across the organization.

7.6 Reporting & Insights

The reporting module generates detailed analytics to support decision-making.

It includes:

- Incident trend analysis over time
- Response time evaluation
- Frequency of different attack types
- Performance metrics of response teams

These insights help improve future cybersecurity strategies.

INCIDENT RESPONSE WORKFLOW

8. Incident Response Workflow

The workflow ensures a standardized process for managing cybersecurity incidents:

1. Threat Detection
2. Alert Generation
3. Incident Recording
4. Severity Classification
5. Assignment to Response Team
6. Investigation and Analysis
7. Containment of Threat
8. System Recovery
9. Incident Closure
10. Post-Incident Review and Learning

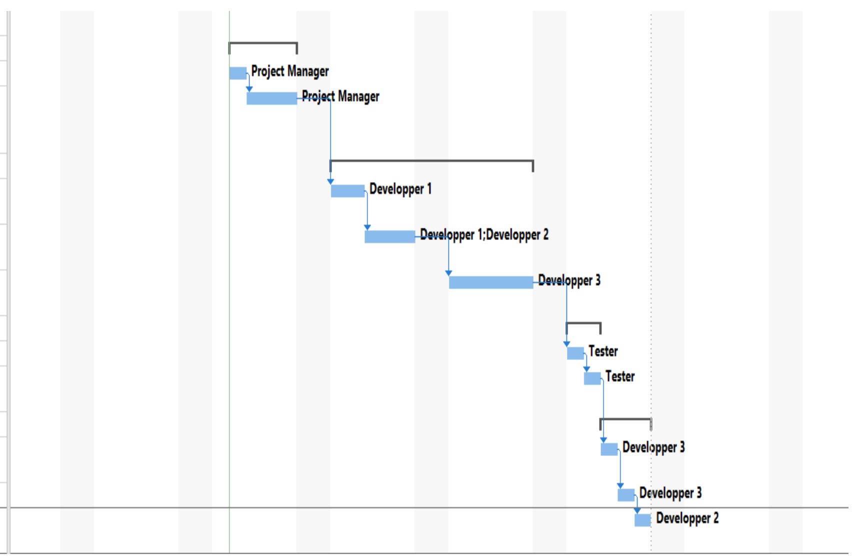
This structured approach ensures no step is missed during incident handling.

PROJECT PLANNING OVERVIEW

9. Project Planning Overview

The project planning phase defines how the Cybersecurity Incident Response Dashboard is developed, structured, and delivered within a limited time frame. It ensures that all tasks are organized, time-bound, and aligned with project objectives.

Planification	**4 jours**	**Mar 26/05/2**	**Ven 29/05/2**	
Needs analysis	1 jour	Mar 26/05/2	Mar 26/05/2	
drafting the technical document	3 jours	Mer 27/05/26	Ven 29/05/26	3
Development	**10 jours**	**Lun 01/06/2**	**Ven 12/06/2**	
Implementing the interfaces	2 jours	Lun 01/06/26	Mar 02/06/26	4
Implementing the DashBoard	3 jours	Mer 03/06/26	Ven 05/06/26	6
Implementing the backend	5 jours	Lun 08/06/26	Ven 12/06/26	7
Testing	**2 jours**	**Lun 15/06/2**	**Mar 16/06/2**	
Performing tests	1 jour	Lun 15/06/2	Lun 15/06/2	8
Performing simulations	1 jour	Mar 16/06/26	Mar 16/06/26	10
Deployment	**3 jours**	**Mer 17/06/2**	**Ven 19/06/2**	
Building the software	1 jour	Mer 17/06/26	Mer 17/06/26	11
Testing	1 jour	Jeu 18/06/2	Jeu 18/06/2	13
Deployment to GitHub	1 jour	Ven 19/06/26	Ven 19/06/26	14



CYBERSECURITY THREATS

10. Cybersecurity Threat Overview

SMEs are exposed to various cybersecurity threats that continue to evolve in complexity.

Major Threats:

- **Ransomware Attacks:** Encrypts organizational data and demands ransom for recovery
- **Phishing Attacks:** Fraudulent emails designed to steal credentials
- **Malware Infections:** Malicious software that damages systems or steals data
- **Insider Threats:** Security risks originating from employees or internal users
- **Data Breaches:** Unauthorized access to sensitive information
- **DDoS Attacks:** Overloads systems to disrupt services

Understanding these threats is essential for building effective defense mechanisms.

BENEFITS TO SMEs

11. Benefits to SMEs

The implementation of a cybersecurity incident response dashboard provides several advantages:

- Faster detection and response to cyber incidents
- Improved operational efficiency
- Enhanced awareness of security risks
- Reduced financial and operational losses
- Better compliance with cybersecurity regulations
- Centralized visibility of all security activities

These benefits collectively strengthen the organization's cybersecurity posture.

CONCLUSION

12. Conclusion

The **Cybersecurity Incident Response Dashboard for SMEs** is designed to address one of the most significant challenges faced by modern organizations: the ability to effectively detect, manage, and respond to cybersecurity incidents. As cyber threats continue to increase frequency and sophistication, SMEs are becoming attractive targets due to their limited security resources and lack of dedicated cybersecurity teams.

This project proposes a centralized dashboard that brings together incident monitoring, alert management, risk assessment, incident tracking, reporting, and response guidance into a single platform. By providing real-time visibility into security events, the dashboard enables organizations to identify threats earlier, prioritize incidents based on severity, and coordinate response activities more efficiently.

One of the key strengths of the proposed solution is its ability to standardize the incident response process. Instead of relying on manual communication and disconnected tools, organizations can follow a structured workflow that guides users from threat detection to incident closure. This not only reduces response times but also minimizes the risk of human error during critical security events.

The dashboard also supports data-driven decision-making through analytical reports and visual insights. Security teams and managers can monitor incident trends, evaluate response performance, identify recurring vulnerabilities, and make informed decisions about future cybersecurity investments. Historical incident data further contributes to continuous improvement by helping organizations learn from past experiences and strengthen their defenses.

From a business perspective, implementing such a system can lead to significant benefits, including reduced operational downtime, lower financial losses, improved compliance with cybersecurity regulations, and increased customer trust. A well-managed incident response capability demonstrates an organization's commitment to protecting sensitive information and maintaining business continuity.

Furthermore, the project highlights the importance of cybersecurity awareness and preparedness in today's digital environment. Technology alone cannot eliminate cyber risks; however, combining effective tools, structured processes, and informed personnel can significantly reduce the impact of attacks.

In conclusion, the Cybersecurity Incident Response Dashboard provides a practical, scalable, and cost-effective solution for SMEs seeking to improve their cybersecurity posture. By enhancing visibility, coordination, and response efficiency, the system strengthens organizational resilience and helps businesses operate more securely in an increasingly complex threat landscape. The project demonstrates how a centralized incident management platform can play a vital role in protecting digital assets, supporting business continuity, and ensuring long-term cybersecurity readiness.